GURUNADH MALLI

Founding AI Engineer · Generative AI & Agentic AI

PROFESSIONAL SUMMARY

Founding AI Engineer building production Generative AI and Agentic AI systems, from multi-agent workflows and advanced RAG to Model Context Protocol (MCP) integrations. Delivered OAuth 2.1-secured MCP connectors that bring product modules into ChatGPT and Claude Desktop, and agent harnesses with hooks and skills for safe, controlled tool execution. Skilled in LangChain, LangGraph and LangSmith observability, hybrid and graph RAG, and multi-LLM orchestration across AWS Bedrock (Claude), Gemini, OpenAI, Cohere and Groq, on a Django, FastAPI, React/TypeScript and AWS serverless stack with GitHub Actions CI/CD.

TECHNICAL SKILLS

Generative AI: AWS Bedrock (Claude Sonnet, Haiku, Titan), OpenAI, Google Gemini, Cohere, Groq, Prompt Engineering, Fine-tuning

Agentic AI: LangChain, LangGraph, Google ADK, Multi-Agent Systems, Tool Calling, Agent Harness, Hooks, Skills, Human-in-the-Loop, Agent Memory

MCP & Security: MCP Servers & Clients, Remote MCP Connectors (ChatGPT, Claude Desktop), OAuth 2.1, OAuth2, JWT, RBAC

Observability: LangSmith (tracing, prompt versioning), RAGAS

RAG: Hybrid Search (Dense + BM25), Re-ranking, Metadata Filtering, Advanced Chunking, Query Rewriting, Multi-Query, Contextual Compression, GraphRAG

Vector & Graph DBs: Pinecone, Qdrant, Weaviate, Milvus, pgvector, FAISS, Neo4j

Languages & Frameworks: Python, JavaScript, SQL, HTML, CSS, Django, FastAPI, React

AWS Cloud: Bedrock, EC2, S3, Lambda, Lambda Durable Functions, SQS, SNS, SES, RDS, CloudWatch, Cost Explorer, SageMaker (Fundamentals)

DevOps & Tools: CI/CD with GitHub Actions, Git, Google Cloud Tasks, Playwright, Postman, Jira, Jupyter, Agile/Scrum

Machine Learning: Supervised Learning, Unsupervised Learning, Scikit-learn, Pandas, NumPy, CNN, RNN, Transformers, OCR

Databases & Integrations: PostgreSQL, MySQL, Microsoft Graph API, Meta, LinkedIn

EXPERIENCE

Founding AI Engineer – Generative AI

Oct 2025 – Present

Sagara Global · Hyderabad, India

TradeEximOS – AI Sales, CRM & Marketing Automation Platform

Python · Django · FastAPI · LangChain · LangGraph · LangSmith · MCP · AWS Bedrock (Claude) · Pinecone · PostgreSQL · Playwright

- Developed a multi-tenant AI Sales Intelligence, CRM and Marketing Automation platform combining LLM assistants, RAG and agentic workflows to automate lead management, opportunity analysis, content generation and campaign execution.
- Shipped the Marketing module as a **remote MCP connector for ChatGPT and Claude Desktop**, secured with **OAuth 2.1** (PKCE, dynamic client registration, scoped tokens), so users run campaigns, strategies and content workflows from their AI assistant.
- Designed an **agent harness with hooks and skills**: pre/post tool-call hooks for validation, guardrails, permission checks and audit logging, plus modular skills packaging instructions and tools per sales and marketing task.
- Implemented **LangSmith observability** across all agents: tracing of runs, tool calls, tokens and latency, plus prompt versioning to debug agents and catch regressions before release.

- Built a LangGraph sales assistant on AWS Bedrock (Claude Sonnet & Haiku) with tool-based retrieval of leads, opportunities and interactions, delivering next-best actions, multilingual email drafting and scheduling via Microsoft Graph API.
- Built a production RAG service with Amazon Titan embeddings, hybrid retrieval (semantic + BM25), metadata filtering, re-ranking and OCR, with FastAPI services for ingestion, chunking and Pinecone storage.
- Separated business logic (Django) from AI services (FastAPI for RAG, LLM workflows and content generation); added OAuth2/JWT auth and RBAC (admin, sales, marketing) across tenants on AWS Bedrock, S3 and RDS.
- Built OAuth2-secured integrations with Outlook, Instagram, Facebook and LinkedIn, with asynchronous email and social scheduling on Google Cloud Tasks.

Emboss – AI-Powered HR Management Platform

Python · Django · SQL · React · TypeScript · LangChain · LangGraph · Google Gemini

- Developed role-based automation and workflow agents with LangChain and LangGraph on Google Gemini, calling multiple HR tools to cut manual HR effort and improve operational accuracy.
- Built scalable backend APIs with Python, Django and SQL for HR workflows, with a React and TypeScript frontend for employees, managers and HR admins.
- Automated the employee lifecycle: onboarding, documents, attendance, leave and payroll.
- Integrated AI features including LLM-based resume parsing into structured profiles and smart, context-aware notifications.

AI/ML Intern

Mar 2025 – Oct 2025

Ignitz Solutions Pvt. Ltd. · Hyderabad, India

Enterprise RAG Chatbot & Executive Intelligence Assistant

Python · FastAPI · LangChain · LangGraph · Cohere · Groq · Gemini · Qdrant · FAISS · PostgreSQL · Neo4j · RAGAS

- Built an **end-to-end RAG chatbot** over enterprise construction documents and data, orchestrating **Cohere, Groq and Gemini** LLMs with complexity-based routing and automatic fallback for reliability and cost.
- **Ingestion & chunking:** parsed PDF, DOCX, spreadsheets and scans (OCR) with metadata enrichment; recursive, semantic and parent-child chunking with tuned overlap.
- **Retrieval:** query rewriting and multi-query expansion; Cohere embeddings in Qdrant and FAISS plus BM25 fused with Reciprocal Rank Fusion; metadata and self-query filtering; Cohere Rerank and contextual compression.
- **Generation:** cited, grounded answers with conversational memory, FastAPI streaming, hallucination and off-topic guardrails, and semantic caching.
- **Evaluation:** tuned chunk size, top-k and re-ranking with RAGAS (faithfulness, answer relevancy, context precision and recall); delivered executive summaries and trend insights for C-suite leaders.

EDUCATION

Bachelor of Electrical and Electronics Engineering 2022 – 2025

Audisankara Institute of Technology, Gudur, Andhra Pradesh · CGPA 8.73 / 10